

AI Agent Factory

Foundations Course — agentfactory.panaversity.org

PREPARED FOR
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MODULE 2 — AI PROMPTING IN 2026

Total Questions: 30 Parts: 4 Concepts Covered: 13 Marks Each: 4 Total Marks: 120

PART 1 HOW AI KNOWS THINGS (CONCEPTS 1–3)

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CONCEPT 1 — NOVICE VS POWER USER

Zara wants to buy a new car. She simply types "suggest a good car" into the AI. Meanwhile, Ahmed uploads his insurance quotes, dealer quotes, and cost-of-ownership spreadsheets with specific constraints. Whose approach will get a more useful, structured result?

- A Zara — because simpler questions get faster answers
- B Ahmed — because power users provide files, context, and constraints for better results**
- C Both will get the same result since AI is intelligent
- D Zara — AI prefers shorter prompts

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CONCEPT 2 — PRETRAINED KNOWLEDGE

A doctor asks an AI about the latest unpublished clinical trial results from her hospital's internal research database. The AI confidently gives an answer. Should the doctor trust this response?

- A Yes — AI has access to all medical databases**

- B** No — AI cannot know private, unpublished data; confidence is not a signal of correctness
- C** Yes — AI's reasoning mode compensates for missing data
- D** Only if the doctor uses Claude instead of ChatGPT

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CONCEPT 2 — PRETRAINED KNOWLEDGE

Which of these topics will AI answer **MOST** reliably from its pretrained knowledge?

- A** Regional history of a small village in rural Sindh
- B** Internal HR policy of a private company
- C** Popular Python programming concepts
- D** Yesterday's unpublished stock earnings

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CONCEPT 3 — THREE RETRIEVAL MODES

Bilal asks AI: "What happened in the Pakistan election last week?" and gets an answer in 3 seconds from training data alone. But the election was recent. Which retrieval mode did the AI use, and what was the risk?

- A** Pretrained mode — risk of missing recent information beyond knowledge cutoff
- B** Web search mode — risk of citing outdated sources
- C** Deep research mode — too slow for this kind of query
- D** Reasoning mode — AI explored multiple approaches

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CONCEPT 3 — THREE RETRIEVAL MODES

A research analyst needs a comprehensive, structured report on "Global AI regulations in 2026" by scanning dozens of sources. This will take several minutes. Which retrieval mode is she using?

- A** Pretrained mode
- B** Web search mode
- C** Deep research mode
- D** Reasoning / thinking mode

PART 2 TALKING TO AI WELL (CONCEPTS 4-7)

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CONCEPT 4 — CONTEXT IS THE GAME

Hina has been chatting with an AI for 3 hours in the same window. She notices the AI has started forgetting what she said earlier in the conversation. What is the most likely cause?

- A** The AI model was updated during the conversation
- B** Her internet connection is slow

- ☒ C Long conversation "compaction" — older turns summarize into brief paragraphs, losing specificity
- ☐ D The AI ran out of pretrained knowledge

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CONCEPT 4 — CONTEXT IS THE GAME

A modern AI model can hold approximately how many words in its context window — enough for how many Harry Potter books?

- ☐ A 50,000 words — about half a Harry Potter book
- ☐ B 200,000 words — about 1 Harry Potter book
- ☒ C 750,000+ words — roughly 4–5 Harry Potter books
- ☐ D 10,000 words — a short article

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CONCEPT 4 — CONTEXT IS THE GAME

Sara uploads her project brief once into a Claude "Project" and then has 10 different conversations about that project — without re-uploading the brief each time. What feature is she using?

- ☐ A Web search mode
- ☒ B Projects / Notebooks — front-load context once instead of repeating per chat
- ☐ C Deep research mode
- ☐ D System prompt override

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CONCEPT 5 — REASONING / THINKING HARD

Usman has a very complex legal case analysis to do. He types "think carefully before answering" in his prompt. According to Module 2, what does this do?

- A Makes AI respond more slowly for style purposes
- B Switches AI to web search mode
- C Invokes the reasoning/thinking mode — AI explores multiple approaches before answering
- D Reduces hallucinations by turning off creativity

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CONCEPT 5 — REASONING

According to a 2025 METR study, frontier AI models doubled their task-completion capability approximately how often?

- A Every 2 years
- B Every 12 months
- C Every 7 months
- D Every 3 months

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CONCEPT 6 — SYCOPHANCY NEUTRALIZATION

Fatima asks AI: "Can you find evidence that my business plan is brilliant?" The AI enthusiastically agrees and praises the plan. This is a classic example of:

- A Pretrained knowledge bias
- B Context window overflow
- C AI sycophancy — AI trained to agree rather than give honest evaluation**
- D Deep research mode output

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CONCEPT 6 — SYCOPHANCY NEUTRALIZATION

To get an honest evaluation of his marketing strategy, Khalid should ask the AI:

- A "Tell me why this strategy is perfect."
- B "Find proof that my strategy will work."
- C "Evaluate this strategy — list the strongest arguments for AND against it, then score it 1–10."**
- D "Is my strategy good? Yes or no."

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CONCEPT 7 — BRAINSTORM-ITERATE LOOP

What is the **CORRECT** order of the Brainstorm-Iterate Loop — the single highest-leverage AI habit?

- A Get full draft → give feedback → iterate
- B Ask one question → accept first answer → move on
- C Load context → demand 3–5 options → give explicit feedback → iterate 2–3 rounds → expand chosen option → grade 1–10 → iterate until ~9.5**

D Upload files → ask AI to write full document → accept output

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CONCEPT 7 — BRAINSTORM-ITERATE LOOP

Imran asks AI for a product name. AI gives one option. Imran says "great!" and uses it. According to the Brainstorm-Iterate Loop, what did Imran do wrong?

A He should have used web search mode

B He should have demanded 3–5 options first, given explicit feedback, iterated rounds before expanding

C He should have asked in a different language

D He accepted too late in the conversation

PART 3 BEYOND TEXT — MULTIMODAL, APPS & DATA (CONCEPTS 8–11)

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CONCEPT 8 — MULTIMODAL IMAGES

A designer uploads an image of a whiteboard sketch to AI and asks for the text content. Then she asks AI to count exactly 347 tiny dots in the image. Which task will AI do BETTER?

A Reading the whiteboard text — AI is strong at whiteboard content and handwritten text

B Counting the 347 dots — AI is accurate at counting small objects

C Both tasks with equal accuracy

D Neither — AI cannot process images

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CONCEPT 8 — DESIGNER DIAGRAM WORKFLOW

Ali wants to create a professional-quality diagram using AI. According to the Module 2 workflow, what is the correct sequence?

- A Ask AI to draw image → download → use directly
- B Ask Claude to create SVG → render to PNG → paste in ChatGPT/Gemini for redesign → iterate 3–4 rounds**
- C Upload sketch → ask AI to trace it → export as PDF
- D Use deep research mode to find a template online

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CONCEPT 8 — AUDIO

A podcaster uploads a recording of 5 different speakers discussing a topic simultaneously. She asks AI to identify who said what. What should she expect?

- A Perfect identification — AI excels at multi-speaker scenarios
- B Poor results — speaker identification is weak beyond 2 speakers**
- C AI will refuse the task entirely
- D Results depend on internet speed

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CONCEPT 9 — BUILDING APPS WITH ONE PROMPT

Nadia asks Claude: "Build me a bill splitter app where I enter total amount and number of friends, and it shows each person's share." Claude builds this directly in chat. What is this feature called in Claude?

A Canvas

B Artifacts

C Skillpack

D OpenCode

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CONCEPT 9 — BUILDING APPS

A developer asks AI to build a multiplayer online game with real-time server connections. According to Module 2, this is an example of:

A A task AI can easily do with Artifacts

B Something that works in Canvas mode

C Something that does NOT work — AI apps can't handle multiplayer systems and real-time external integrations

D A task for deep research mode

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CONCEPT 10 — DATA ANALYSIS

Rehan uploads his shop's 3-year sales spreadsheet to AI and asks for trend analysis. The AI gives a beautiful-looking answer but Rehan is unsure if it actually ran code. What should Rehan have demanded?

A A PDF report instead of plain text

B "Write and run code to answer this. Show me the code you ran." — explicitly demand code execution

C A second opinion from another AI model

D Use pretrained mode instead

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CONCEPT 11 — DESKTOP APPS & PERMISSIONS

Sana gives an AI desktop tool full access to her entire hard drive and asks it to reorganize all her files. The AI deletes a folder. She tries to recover from Recycle Bin but cannot. Why?

A The Recycle Bin was full

B She used the wrong AI tool

C Files deleted by AI desktop tools do NOT go to the Recycle Bin — they are permanently deleted

D She needed to enable version control first

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CONCEPT 11 — SAFE WORKFLOW

When using an AI desktop tool for the first time, what is the SAFEST workflow according to Module 2?

A Give full drive access and let AI work autonomously

B Ask AI to execute immediately without reviewing

C Tell task → request PLAN (not action) → review and edit plan → approve execution; scope permissions to smallest folder needed

D Only use AI tools after backing up to the cloud automatically

PART 4 WORKING SAFELY & CHOOSING TOOLS (CONCEPTS 12–13)

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CONCEPT 12 — COST & MODEL SELECTION

A creative director needs to generate 50 different video clips to test ideas. She knows each video costs a few dollars and takes several minutes. According to Module 2, what strategy should she use?

- A Generate all 50 videos and pick the best
- B Use audio mode instead — it's cheaper
- C Invest more time in writing detailed text prompts using text AI first — iteration cost shapes strategy
- D Use pretrained mode which is free

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CONCEPT 12 — MODEL STRENGTHS

A journalist needs to pull the latest news from multiple live websites, synthesize them, and connect them with Google Drive. Which AI tool is currently BEST suited for this?

- A Claude — best for reasoning and document understanding
- B ChatGPT — best for image generation and voice
- C Gemini — best for web search synthesis, deep research, and Google integration
- D DeepSeek — best for all tasks

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CONCEPT 12 — MODEL STRENGTHS

A software engineer needs an AI to carefully read a 200-page legal contract, reason through its clauses, and write precise code to extract specific data

from it. Which AI is currently **STRONGEST** for this task?

- A** Claude — strongest in reasoning, document understanding, and careful writing
- B** ChatGPT — strongest for image generation
- C** Gemini — strongest for Google integration
- D** All three are identical in all tasks

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CORE THEME — THE FUNDAMENTAL INSIGHT

According to Module 2, "almost every advanced prompting technique is one of two moves." What are those two moves?

- A** Asking shorter questions and using bullet points
- B** Using Claude for coding and ChatGPT for images
- C** Getting the **RIGHT** context **IN** — or keeping the **WRONG** context **OUT**
- D** Using pretrained mode and deep research mode alternately

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CORE THEME — THE BRIEFING PRINCIPLE

Module 2 says to "brief AI like a smart new colleague." What does this mean in practice?

- A** Always use formal language with AI
- B** Keep prompts under 50 words

- ☐ C Ask: "Would a smart person have enough information to do this job well?" If not — provide more context
- ☐ D Address AI with "Dear AI" to trigger better responses

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NON-OBVIOUS CHANGES SINCE 2022-23

Which of the following is a MAJOR change in AI capabilities that happened AFTER 2022–2023 that Module 2 highlights?

- ☐ A AI learned to speak in 5 languages
- ☒ B Context windows grew ~1000x, enabling hour-long task completion and transparent code execution
- ☐ C AI became free for all users worldwide
- ☐ D AI stopped making any errors

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CONCEPT 12 — COSTS

Rank these AI output types from CHEAPEST to MOST EXPENSIVE per generation:

- ☐ A Video → Image → Speech → Text
- ☐ B Image → Text → Speech → Video
- ☒ C Text → Speech → Image → Video (Text cheapest, Video most expensive)
- ☐ D Speech → Image → Text → Video

HABIT HIERARCHY

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According to the Module 2 Habit Hierarchy, when should you start a NEW conversation with AI instead of continuing the same one?

- A

After every 5 messages
- B

When the AI gives a wrong answer
- C

When topics shift — to avoid context contamination from unrelated previous conversation
- D

After uploading more than 3 files

ANSWER KEY — ALL 30 QUESTIONS

Q1	Q2	Q3	Q4	Q5
B	B	C	A	C
Q6	Q7	Q8	Q9	Q10
C	C	B	C	C
Q11	Q12	Q13	Q14	Q15
C	C	C	B	A
Q16	Q17	Q18	Q19	Q20
B	B	B	C	B
Q21	Q22	Q23	Q24	Q25
C	C	C	C	A
Q26	Q27	Q28	Q29	Q30
C	C	B	C	C

